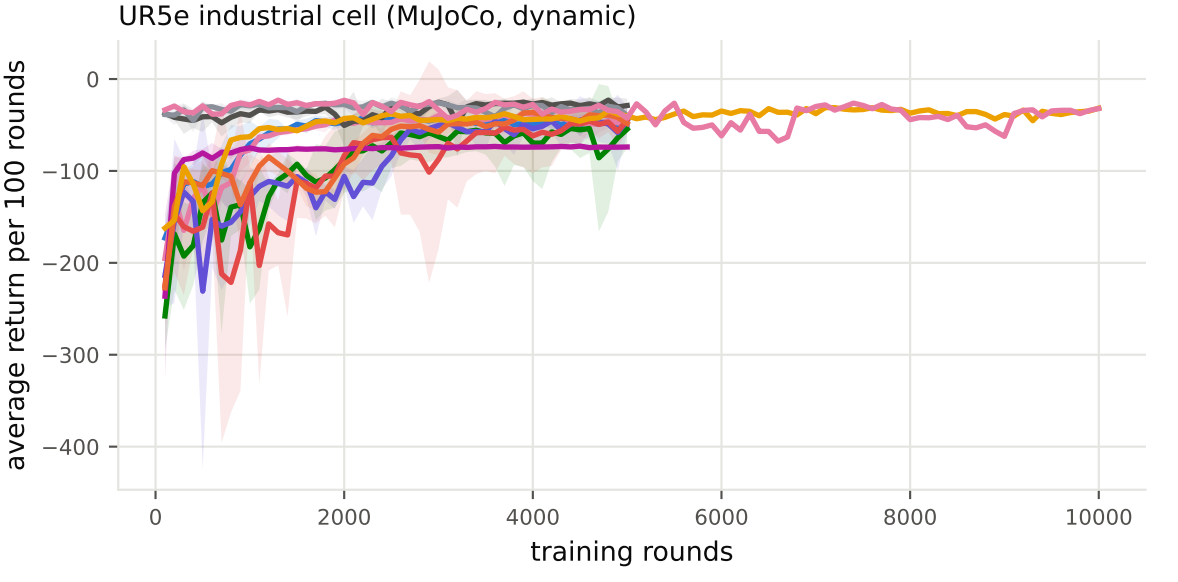
UR5e industrial cell (MuJoCo, dynamic)
average return per 100 rounds
0
−100
−200
−300
−400
0
2000
4000
6000
8000
10000
training rounds
DDPG (paper)
+ TD3
+ HER
+ OBL
+ APF shaping
+ APF prior (residual)
+ CBF shield
CEM-RL (population) on TD3+HER
+ HER (k = 1)
DDPG (paper), 10 000 rounds
TD3+HER+APF+shield, 10 000 rounds